

FindSkill **a freelancing platform**

A Minor Project Report
submitted in partial fulfillment of the requirements for
the award of the degree of

Bachelor of Engineering

in

Artificial Intelligence and Data Science

By

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY
HYDERABAD – 500075
DECEMBER 2023



**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
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HYDERABAD – 500075**

INSTITUTE VISION

“To be the center of excellence in technical education and research”.

INSTITUTE MISSION

“To address the emerging needs through quality technical education and advanced research”.

DEPARTMENT VISION

”To be a globally recognized center of excellence in the field of Artificial Intelligence and Data Science that produces innovative pioneers and research experts capable of addressing complex real-world challenges and contributing to the socio-economic development of the nation.”

DEPARTMENT MISSION

1. To provide cutting-edge education in the field of Artificial Intelligence and Data Science that is rooted in ethical and moral values.
2. To establish strong partnerships with industries and research organizations in the field of Artificial Intelligence and Data Science, and to excel in the emerging areas of research by creating innovative solutions.
3. To cultivate a strong sense of social responsibility among students, fostering their inclination to utilize their knowledge and skills for the betterment of society.
4. To motivate and mentor students to become trailblazers in Artificial Intelligence and Data Science, and develop an entrepreneurial mindset that nurtures innovation and creativity.



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PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

Graduates of AI & DS will be able to:

1. Adapt emerging technologies of Artificial Intelligence & Data Science and develop state of the art solutions in the fields of Manufacturing, Agriculture, Health-care, Education, and Cyber Security.
2. Exhibit professional leadership qualities to excel in interdisciplinary domains.
3. Possess human values, professional ethics, application-oriented skills, and engage in lifelong learning.
4. Contribute to the research community to meet the needs of public and private sectors.

PROGRAM SPECIFIC OUTCOMES (PSOs)

After successful completion of the program, students will be able to:

1. Exhibit proficiency of Artificial Intelligence and Data Science in providing sustainable solutions by adapting to societal, environmental and ethical concerns to real world problems.
2. Develop professional skills in the thrust areas like ANN and Deep learning, Robotics, Internet of Things and Big Data Analytics.
3. Pursue higher studies in Artificial Intelligence and Data Science in reputed Universities and to work in research establishments.



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PROGRAM OUTCOMES

1. **Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization for the solution of complex engineering problems
2. **Problem analysis:** Identify, formulate, review, research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with the society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.



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MINOR PROJECT-I

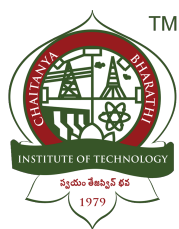
COURSE OBJECTIVES

1. To enable students to learn by doing.
2. To develop capability to analyse and solve real world problems.
3. To inculcate innovative ideas of the students.
4. To impart team building and management skills among students.
5. To instill writing and presentation skills for completing the project.

COURSE OUTCOMES

Upon successful completion of this course, students will be able to:

1. Interpret Literature with the purpose of formulating a project proposal.
2. Plan, Analyse, Design and Implement a project.
3. Find the solution of identified problem with the help of modern Technology and give priority to real time scenarios.
4. Plan to work as a team and to focus on getting a working project done and submit a report within a stipulated period of time.
5. Prepare and submit the Report and deliver presentation before the Departmental Committee.



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CO-PO MAPPING

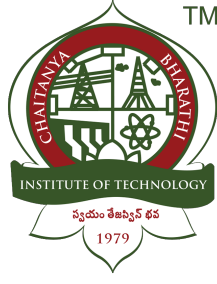
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	3	3	3	3	2	1	2	3	3
CO2	3	3	3	3	3	3	3	2	1	2	3	3
CO3	3	3	3	3	3	3	3	2	-	2	3	3
CO4	2	2	2	3	3	3	3	2	3	3	2	3
CO5	1	2	1	2	3	3	-	-	2	3	-	-

Mapping of Course Outcomes with Program Outcomes

CO-PSO MAPPING

	PSO1	PSO2	PSO3
CO1	2	3	3
CO2	3	3	3
CO3	3	3	3
CO4	2	3	3
CO5	-	3	-

Mapping of Course Outcomes with Program Specific Outcomes



**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
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DECLARATION CERTIFICATE

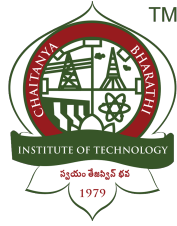
We hereby declare that the project titled **FindSkill, a freelancing platform** submitted by us to the **Artificial Intelligence and Data Science, CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY, HYDERABAD** in partial fulfillment of the requirements for the award of **Bachelor of Engineering** is a bona-fide record of the work carried out by us under the supervision of **Mrs.V.K.Aravinda**. We further declare that the work reported in this project, has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma of this institute or of any other institute or University.

Project Associates

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**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY
HYDERABAD – 500075**

BONAFIDE CERTIFICATE

This is to certify that the project titled **FindSkill, a freelancing platform** is a bonafide record of the work done by

Pasunuri Kathyayini (1601-21-771-088)

Ambekar Tejas (1601-21-771-095)

Cheera Rohan (1601-21-771-099)

in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering in Artificial Intelligence and Data Science** to the **CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY, HYDERABAD** carried out under my guidance and supervision during the year 2023-24. The results presented in this project report have not been submitted to any other university or Institute for the award of any degree.

Mrs.V.K.Aravinda

Guide

Dr.K.Ramana

Head of the Department

Submitted for V Semester Minor-Project viva-voce examination held on _____

Examiner-1

Examiner-2

ABSTRACT

Freelancing is a dynamic and flexible work arrangement where individuals, known as freelancers, offer their skills and services to clients on a project-by-project basis. Freelance, freelancer, or freelance worker, are terms commonly used for a person who is self-employed and not necessarily committed to a particular employer long-term. The modern workforce landscape is characterized by a growing demand for freelance services, as businesses and individuals increasingly seek specialized skills for their projects. However, the existing freelancing platforms often suffer from issues such as high fees, lack of transparency, and difficulties in finding the right talent. These challenges hinder the efficiency of freelance collaborations and limit opportunities for both freelancers and clients. Traditional platforms, including those inspired by Fiverr, have made strides in addressing these concerns but often fall short in providing a holistic solution.

As an enhancement to this model, our proposed freelancing platform represents an updated approach to addressing these challenges. It offers a more user-friendly and cost-effective environment for freelancers to showcase their skills and clients to identify suitable expertise. Through innovative features and a commitment to transparency, our platform aims to bridge the gap between freelancers and clients, fostering a collaborative ecosystem that empowers both parties. This academic project will delve into the architecture and functionality of our platform, highlighting the ways in which it builds upon existing solutions to create a more efficient and inclusive freelance marketplace. The technology stack that we use in order to implement this project MERN (MongoDB, Express.js, React.js, Node.js) stack.

Keywords : Freelancing, MERN, collaborations, efficiency, cost-effective

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We would like to express our deepest gratitude to the following people for guiding us through this course and without whom this project and the results achieved from it would not have reached completion.

Mrs.V.K.Aravinda, Assistant Professor, Department of Artificial Intelligence and Data Science, for helping us and guiding us in the course of this project. Without her guidance, we would not have been able to successfully complete this project. Her patience and genial attitude is and always will be a source of inspiration to us.

Dr.K.Ramana, the Head of the Department, Department of Artificial Intelligence and Data Science, for allowing us to avail the facilities at the department.

We are also thankful to the faculty and staff members of the Department of Artificial Intelligence and Data Science, our individual parents and our friends for their constant support and help.

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CHAPTER 1

INTRODUCTION

The advent of the digital era has revolutionized the way businesses operate, emphasizing the need for a dynamic platform that seamlessly connects skilled freelancers with businesses seeking their expertise. FindSkill, a freelancing platform developed using the MERN stack (MongoDB, Express.js, React.js, Node.js), emerges as a cutting-edge solution designed to address the challenges of traditional hiring processes and foster efficient collaboration in a global marketplace.

1.1 OVERVIEW

FindSkill is a comprehensive online freelancing platform that transcends geographical boundaries, enabling businesses to discover and engage with a diverse pool of talented freelancers. By leveraging the power of the MERN stack, FindSkill ensures a robust and scalable infrastructure, providing a user-friendly experience for both freelancers and businesses alike. The platform facilitates secure transactions, transparent communication, and streamlined project management, offering a centralized hub for professionals and businesses to connect, collaborate, and thrive.

1.2 MOTIVATION

The development of FindSkill is driven by the desire to empower individuals to showcase their skills and connect with diverse opportunities. The project addresses the need for a virtual marketplace that seamlessly connects freelancers with businesses, offering a user-friendly platform fostering collaboration, innovation, and professional growth. FindSkill aims to revolutionize the approach to work in the digital age, providing freelancers a global stage and businesses the tools to find perfect talent, ultimately fostering collaborative success in the freelancing landscape.

1.3 PROBELM STATEMENT

To develop a platform to address the challenge of efficiently matching skilled freelancers with businesses seeking specific expertise while streamlining the hiring process, thereby enhancing collaboration.

CHAPTER 2

SYSTEM REQUIREMENTS

2.1 FUNCTIONAL REQUIREMENTS

a. User Management:

1. User Registration and Authentication:

- Users should be able to create accounts with a valid email address and password.
- Account verification and password recovery mechanisms should be in place.

2. User Profiles:

- Users must be able to create and manage their profiles
- Profiles should contain details such as skills, experience, and portfolio.
- A profile verification process may be implemented for added credibility.

3. Role-Based Access Control:

- Implement different user roles (freelancer, business) with distinct permissions.
- Admins should have special privileges for platform management.

b. Gig and Order Management:

4. Gig Posting:

- Freelancers should be able to post their services as Gigs.
- Gigs with details such as title, description, pricing, portfolio, number of revisions etc.,
- Gigs should be categorized for easy search and discovery.

5. Order Placing:

- Businesses or clients should be able to make orders.
- Communication channels should be available between freelancers and businesses.

6. Order accepting:

- Freelancers should have the ability to accept or reject orders.
- The system should facilitate secure transactions and handle project milestones.

7. Gig Management Dashboard:

- Both freelancers and businesses should have a dashboard to track projects.
- The dashboard should display project status, communications, and transactions.

c. Communication and Collaboration:

8. Real-time Chat:

- Include a real-time messaging system for direct communication
- Notifications for new messages should be provided.

9. File Sharing:

- Enable freelancers and businesses to share files related to projects securely.

d. Reviews and Ratings:

10. Review and Rating System:

- Implement a system for users to leave reviews after project completion.
- Consider a two-way review system for both freelancers and businesses.

e. Financial Transactions:

11. Payment Gateway Integration:

- Integrate a secure payment gateway to facilitate financial transactions.
- Ensure compliance with industry standards for online transactions.

These functional requirements provide a foundation for the development of the FindSkill.

2.2 NON-FUNCTIONAL REQUIREMENTS

1. Performance:

- The platform should support atleast 1000 concurrent users.
- Page load times should be under 3 seconds for optimal user experience.

2. Scalability:

- The system architecture should be scalable to accommodate a growing user base.
- It should seamlessly scale both vertically and horizontally to handle increased traffic.

3. Reliability:

- The system should be able to recover from failures gracefully.
- Regularly perform system health checks and proactive monitoring.

4. Security:

- Ensure data encryption for sensitive information during transmission.
- Implement secure coding practices to prevent common vulnerabilities.

5. Usability:

- The U.I. should be intuitive and accessible, adhering to best web accessibility.
- Conduct usability testing to identify and address user experience issues.

6. Scalability:

- The system should be able to handle an increase in the user base.
- Implement caching mechanisms and optimize database queries for improved scalability.

2.3 SOFTWARE REQUIREMENTS

Frontend:

- *HTML*: Used for structuring web pages.
- *SCSS*: Employed for styling and layout design.
- *JSX*: JavaScript's extension used in React for defining UI components.
- *React*: Framework for building the user interface and managing state efficiently.
- *Vite*: Tool that is a fast and efficient alternative to traditional bundlers for JS projects.

Backend:

- *Node.js*: Server-side JavaScript runtime used to execute server-side code.
- *Express.js*: Framework for Node.js that simplifies the creation of robust APIs.
- *MongoDB*: NoSQL DB for storing & managing data in a flexible, schema-less format.

Database Management:

- *MongoDB Atlas*: Cloud-based service for managing and hosting MongoDB databases.

Development Tools:

- *npm (Node Package Manager)*: Manages dependencies & executes development scripts.

Integrated Development Environment (IDE):

- *Visual Studio Code (VSCode)*: Lightweight and powerful code editor.

Version Control Integration:

- *GitHub*: Web-based platform for version control using Git.

2.4 HARDWARE REQUIREMENTS

Processor: A modern processor like Intel Core i5 or AMD Ryzen with at least two cores is recommended. A faster processor will enhance the performance, especially during server-side processing and handling concurrent requests in the Node.js environment.

RAM: At least 8 GB of RAM is recommended for efficient development and execution of the MERN stack components. More RAM might be needed for handling larger data, especially during server-side operations and database interactions.

Internet Connectivity: A stable and high-speed internet connection is necessary for accessing online resources, collaborating with team members, and deploying the web application to cloud services.

CHAPTER 3

SYSTEM DESIGN OR METHODOLOGY

3.1 SYSTEM ARCHITECTURE

The system design of the FindSkill project encompasses the overall architecture, components, and interactions between different modules. Here's a brief overview:

3.1.1 Technical Framework

1. MERN Stack:

- Utilizes the MERN stack with MongoDB as the database, Express.js as the backend framework, React for the frontend, and Node.js as the runtime environment.

2. Client-Server Architecture:

- Follows a client-server architecture where the React-based client interacts with the Node.js and Express.js backend server.

3.1.2 Frontend Design

3. React Components:

- UI components are designed using React, allowing for modular, reusable, and maintainable code.

4. State Management:

- Utilizes React's state management and props to efficiently handle and pass data between components.

5. Responsive UI:

- Implements a responsive and user-friendly interface using HTML, SCSS, and JSX.

3.1.3 Backend Design

6. Express.js API: - Implements a RESTful API using Express.js, handling requests and responses between the frontend and the database.

7. Middleware: - Uses middleware functions for handling authentication, logging, and other cross-cutting concerns.

8. Database Integration: - Integrates with MongoDB for data storage, utilizing the Mongoose ODM for streamlined database operations.

3.1.4 Data Flow

9. User Authentication:

- Implements secure user authentication and authorization mechanisms using JWT (JSON Web Tokens).

10. Data Validation:

- Validates and sanitizes user inputs to prevent security vulnerabilities and ensure data integrity.

3.1.5 Infrastructure and Deployment

11. MongoDB Atlas:

- Leverages MongoDB Atlas for cloud-based database hosting, ensuring scalability, reliability, and ease of management.

3.1.6 Security

12. HTTPS Encryption:

- Enforces HTTPS for secure data transmission between clients and servers.

13. Input Sanitization:

- Implements input validation and sanitation to mitigate security risks like SQL injection and XSS attacks.

CHAPTER 4

IMPLEMENTATION

Here is an overview of the implementation of the FindSkill project, covering frontend, backend, and data management aspects.

4.1 FRONTEND IMPLEMENTATION

4.1.1 Technologies Used:

- React.js for building the user interface.
- Vite as the build tool for a fast and efficient development workflow.
- JSX and SCSS for component-based styling.

4.1.2 Components and Pages:

1. Navigation Components:

- Header and Footer components for consistent navigation.
- Navbar for easy access to different sections.

2. User Authentication:

- Two types of accounts can be created, client (buyer) and freelancer (seller)
- Login and registration pages with corresponding components.
- Authentication context for managing user sessions.

3. Main contents:

- Freelancer and business dashboards displaying relevant information.
- Project listings, notifications, and user details components.
- Featured section for different domains available such as AI Image Artist, Logo Designer, Web Developer, etc.,

5. Freelancer Dashboard:

- Gig page to create a gig by providing details such as Title, Service offered, Portfolio, Pricing, Submission time, No. of revisions, etc.,
- A page to look at the list of all Gigs created
- A page to track, accept and manage orders made by clients
- A page to manage their own profile
- A chat section to contact and provide updates to the clients

6. Client dashboard:

- Available Gig pages to discover and choose suitable freelancer for their work
- Project details page for in-depth information on specific projects offered by freelancers.

- A section to track and manage the orders placed
- A chat section to contact and stay updated with the freelancers (sellers).

7. Messaging and Collaboration:

- Real-time chat components for communication between freelancers and businesses.
- Collaboration tools for project discussions and updates.
- Review section to provide reviews for each Gig through which service was taken.

8. Payments:

- Payment page to enter card details and proceed to checkout
- Checkout form for confirmation of payment

The following shows the different files created to implement the frontend

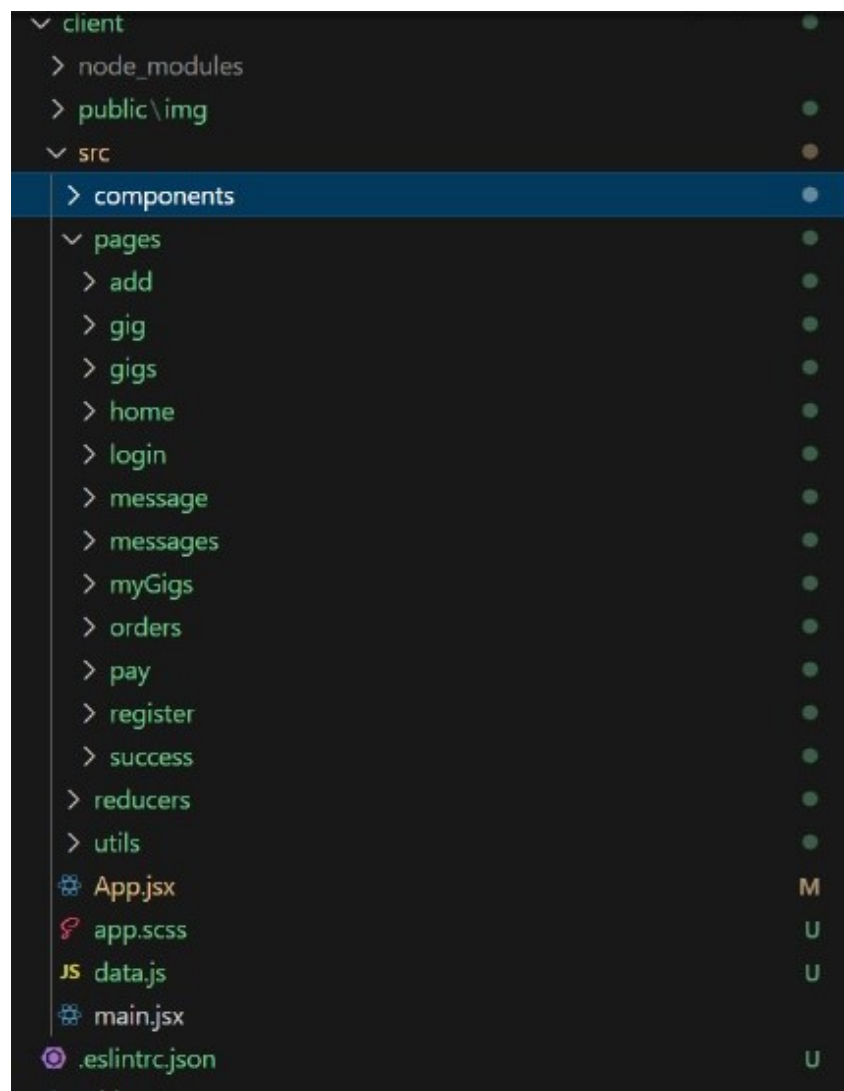


Figure 4.1: Project Files for the Frontend

4.2 BACKEND IMPLEMENTATION

4.2.1 Technologies Used:

- Node.js and Express.js for backend development.
- MongoDB as the NoSQL database for data storage.
- MongoDB Atlas for cloud-based database hosting.

4.2.2 API Endpoints and Routes:

1. User Management:

- User registration and authentication routes.
- Profile management endpoints for updating user information.

2. Gig and Order Management:

- API endpoints for creating, updating, and deleting Gigs and orders.
- Routes for handling freelancer proposals and Gig reviews.

3. Real-time Communication:

- WebSocket implementation for real-time communication through chats.
- Endpoints for handling chat messages and notifications.

The following shows the different files created to implement the backend

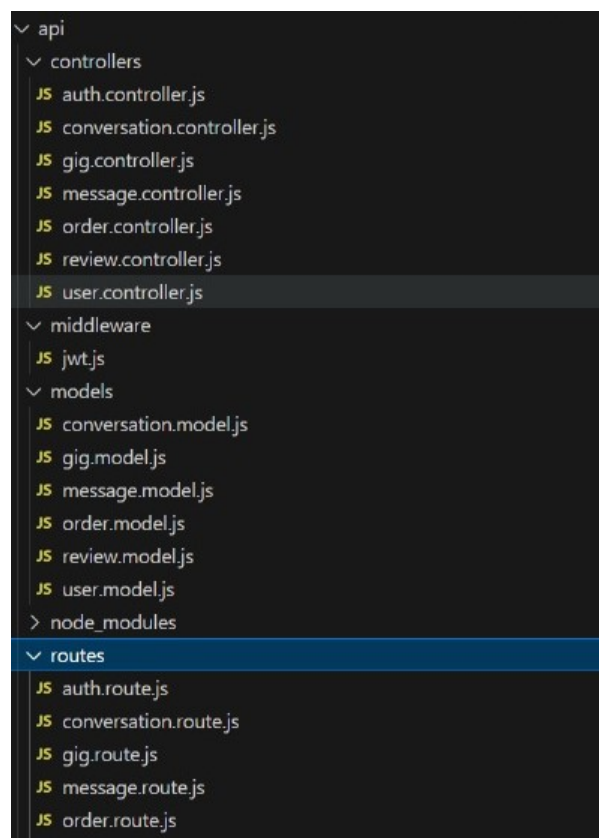


Figure 4.2: Project Files for the Backend

4.3 DATA MANAGEMENT

4.3.1 MongoDB Integration:

- Schemas and models for defining data structures in MongoDB.
- MongoDB Atlas for hosting and accessing data from cloud.

4.3.2 Data Flow:

1. User Authentication Flow:

- Token generation and verification for secure user authentication.
- Middleware functions for ensuring authorized access to protected routes.

2. Data Validation and Sanitization:

- Input validation for preventing invalid data entry.
- Sanitization to protect against security vulnerabilities.

4.3.3 Routes and Navigation:

1. Routing in React:

- React Router for managing navigation within the application.
- Route guards to control access based on user authentication status.

2. Page Components:

- Separate components for each page, promoting modularity and reusability.
- Nested routes for organized navigation structure.

The tables created and the details under each of them are shown below:

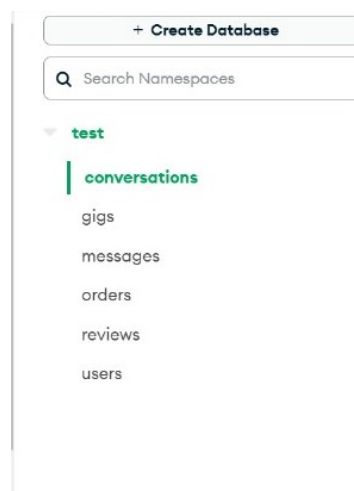


Figure 4.3: The tables created in MongoDB Atlas

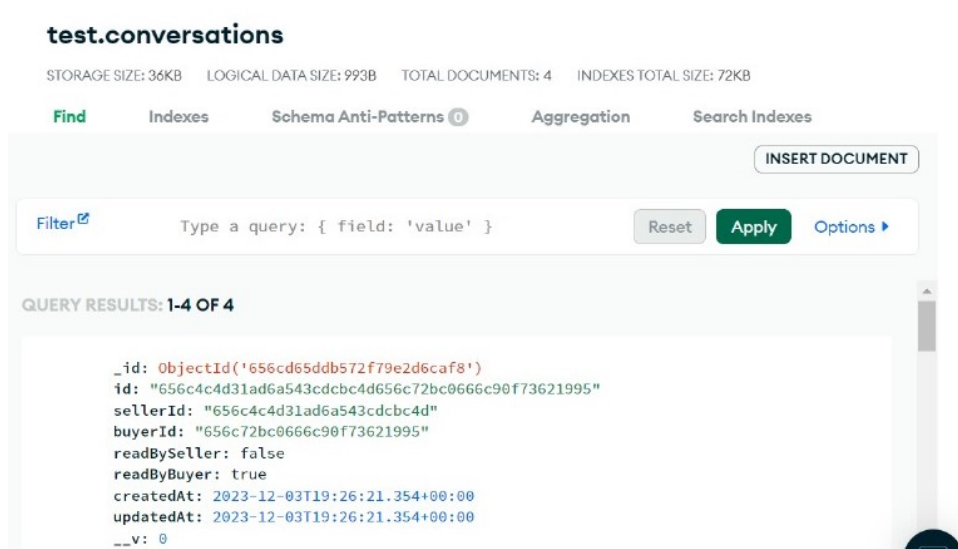


Figure 4.4: The fields of conversation table

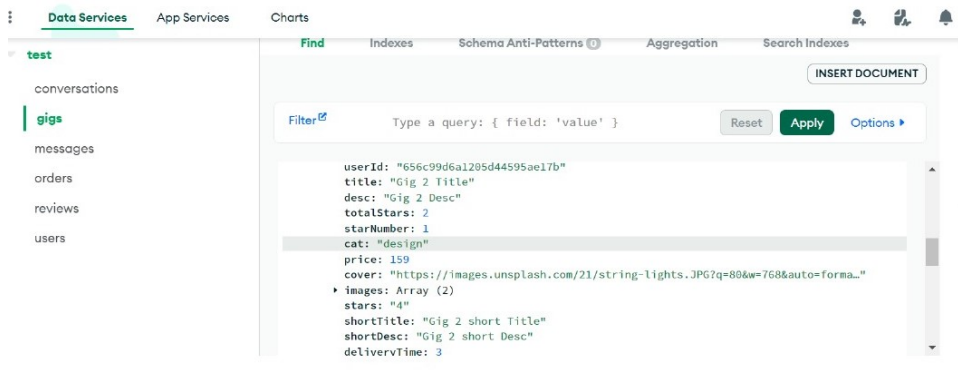


Figure 4.5: The fields of Gigs table

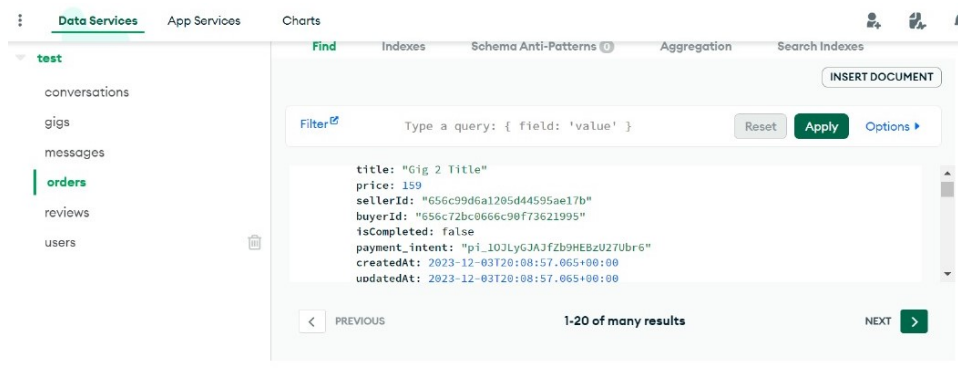


Figure 4.6: The fields of Orders table

CHAPTER 5

TESTING AND RESULTS

The homepage of the FindSkill looks as follows that contains Navigation bar with several sections to easily navigate to, a welcome quote, and a Featured section which contains several cards of different domains available to see Gigs from.

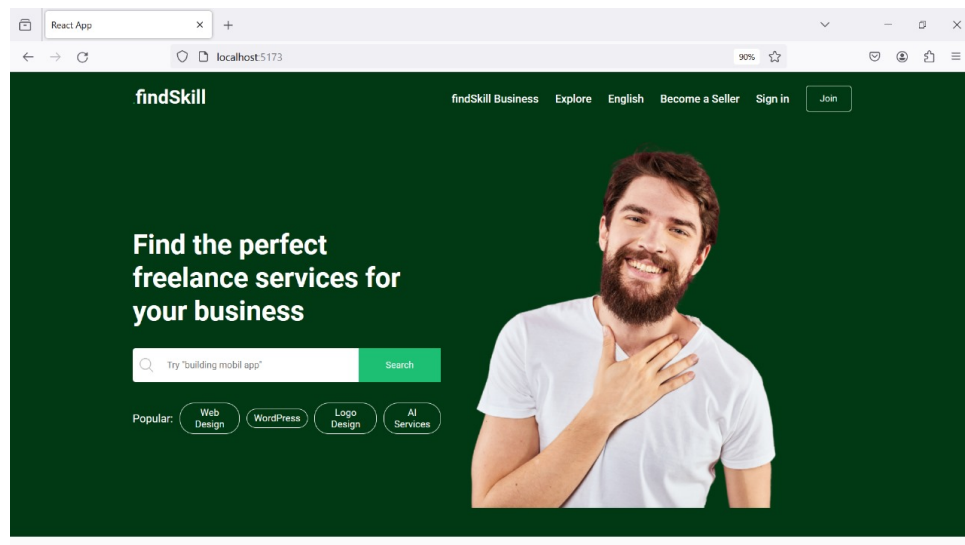


Figure 5.1: The Homepage of the website with Navbar

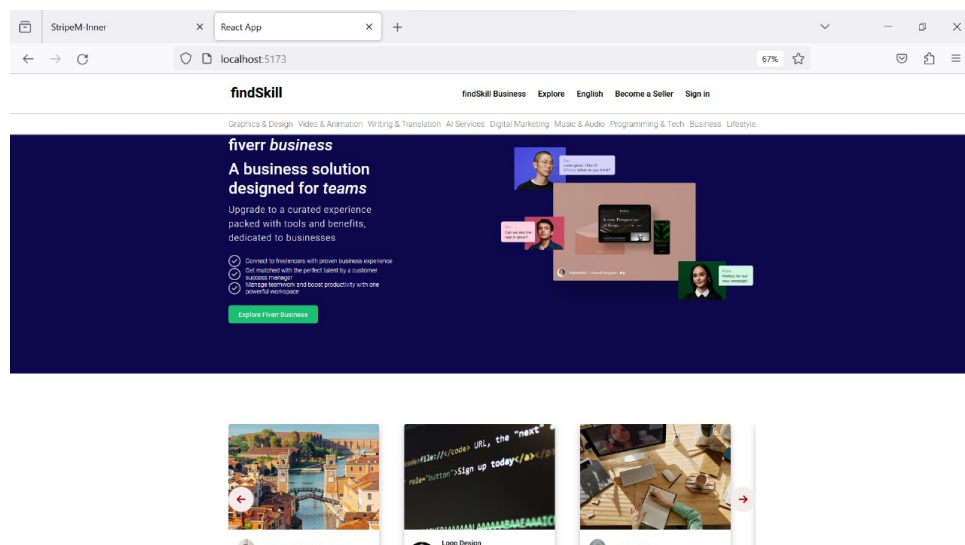


Figure 5.2: The Homepage of the website contd.,

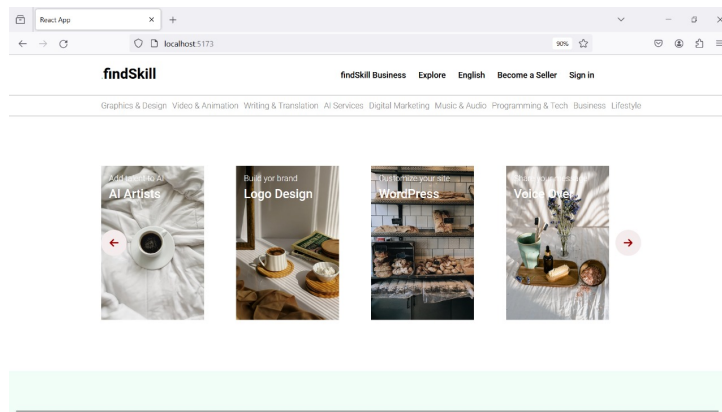


Figure 5.3: The Featured section of the website

After opening a domain card, all Gigs listed in that domain appear as follows

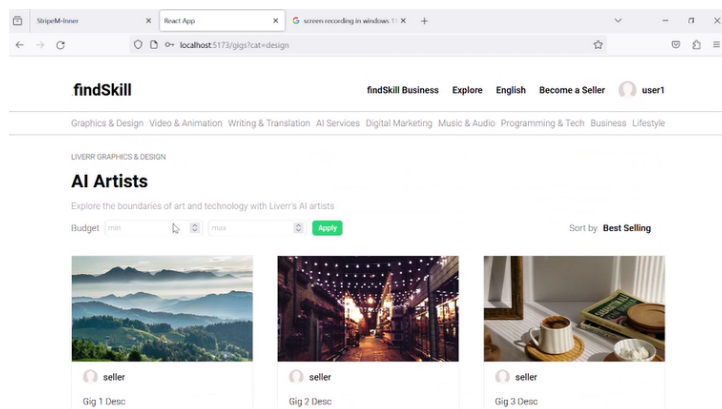


Figure 5.4: The list of all Gigs created by freelancers under that domain

A single Gig page contains all the information of the Gig along with a review section

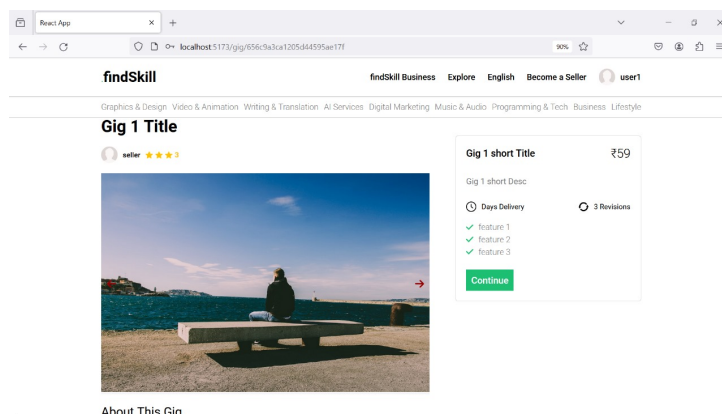


Figure 5.5: The individual page of Gig

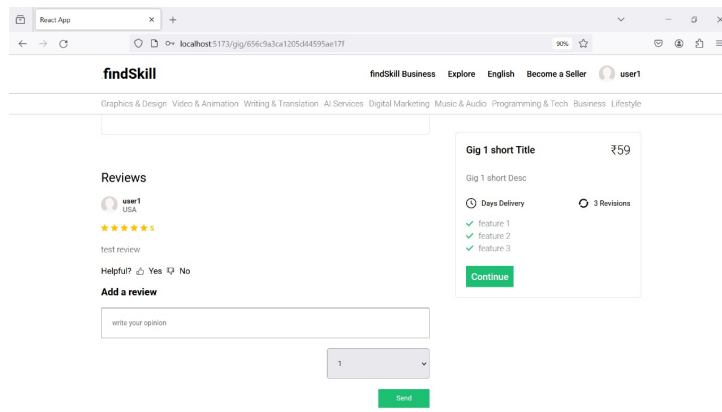


Figure 5.6: The Review section of Gig

Below is the Gig creation page available for seller (freelancer)

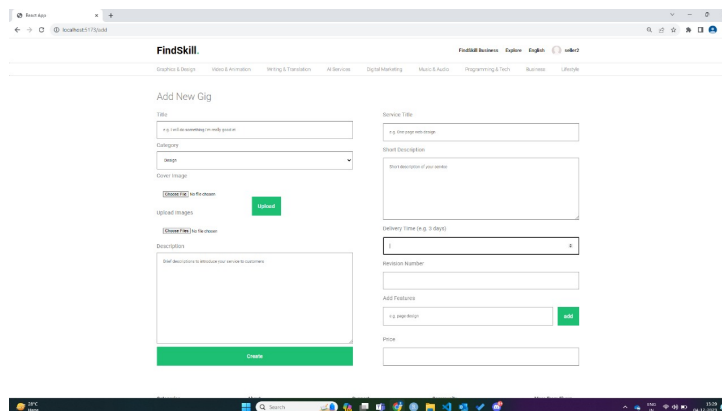


Figure 5.7: The "Create Gig" page

Here is how the list of all Gigs look like

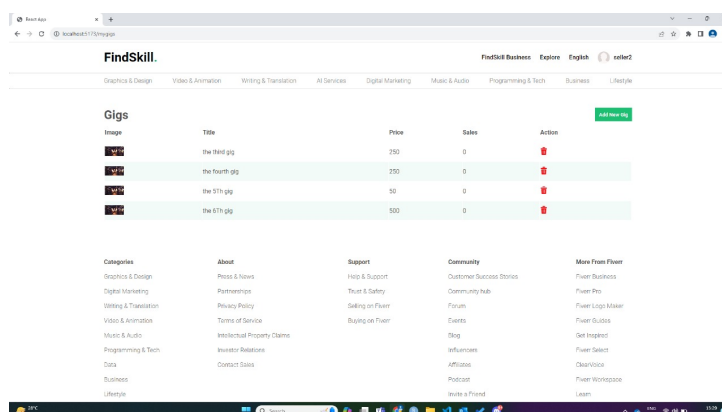
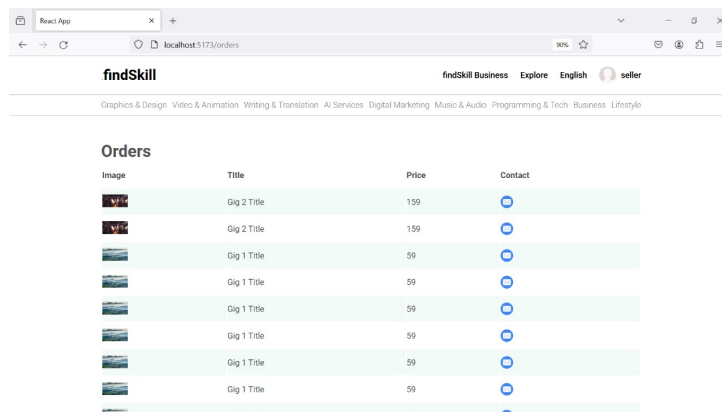


Figure 5.8: List of Gigs created by seller

The list of all orders will be displayed as follows

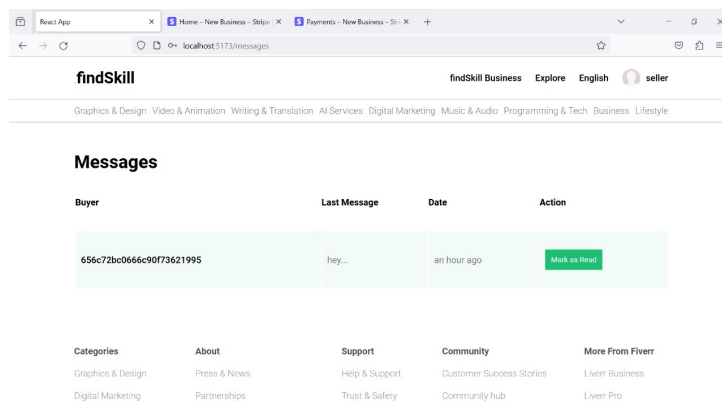


The screenshot shows a web browser window with the URL `localhost:5173/orders`. The page is titled "findSkill" and has a navigation bar with "findSkill Business", "Explore", "English", and a "seller" profile. Below the navigation bar, there is a list of categories: Graphics & Design, Video & Animation, Writing & Translation, AI Services, Digital Marketing, Music & Audio, Programming & Tech, Business, and Lifestyle. The main content area is titled "Orders" and displays a table of orders.

Image	Title	Price	Contact
	Gig 2 Title	199	Contact
	Gig 2 Title	199	Contact
	Gig 1 Title	59	Contact
	Gig 1 Title	59	Contact
	Gig 1 Title	59	Contact
	Gig 1 Title	59	Contact
	Gig 1 Title	59	Contact
	Gig 1 Title	59	Contact

Figure 5.9: List of Orders received by seller

The message section has a list of messages sent to you from several users, they can be marked as read or unread



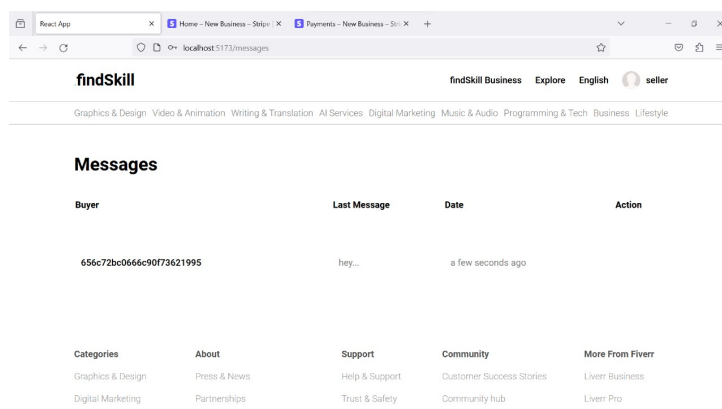
The screenshot shows a web browser window with the URL `localhost:5173/messages`. The page is titled "findSkill" and has a navigation bar with "findSkill Business", "Explore", "English", and a "seller" profile. Below the navigation bar, there is a list of categories: Graphics & Design, Video & Animation, Writing & Translation, AI Services, Digital Marketing, Music & Audio, Programming & Tech, Business, and Lifestyle. The main content area is titled "Messages" and displays a table of messages.

Buyer	Last Message	Date	Action
656c72bc0666c90f73621995	hey...	an hour ago	Mark as Read

Below the messages table, there is a footer section with links to various categories and services:

- Categories:** Graphics & Design, Digital Marketing
- About:** Press & News, Partnerships
- Support:** Help & Support, Trust & Safety
- Community:** Customer Success Stories, Community hub
- More From Fiverr:** Liverr Business, Liverr Pro

Figure 5.10: Inbox section (Message is shown as unread)



The screenshot shows a web browser window with the URL `localhost:5173/messages`. The page is titled "findSkill" and has a navigation bar with "findSkill Business", "Explore", "English", and a "seller" profile. Below the navigation bar, there is a list of categories: Graphics & Design, Video & Animation, Writing & Translation, AI Services, Digital Marketing, Music & Audio, Programming & Tech, Business, and Lifestyle. The main content area is titled "Messages" and displays a table of messages.

Buyer	Last Message	Date	Action
656c72bc0666c90f73621995	hey...	a few seconds ago	

Below the messages table, there is a footer section with links to various categories and services:

- Categories:** Graphics & Design, Digital Marketing
- About:** Press & News, Partnerships
- Support:** Help & Support, Trust & Safety
- Community:** Customer Success Stories, Community hub
- More From Fiverr:** Liverr Business, Liverr Pro

Figure 5.11: Inbox section (Message is shown as read)

The chat page looks as follows

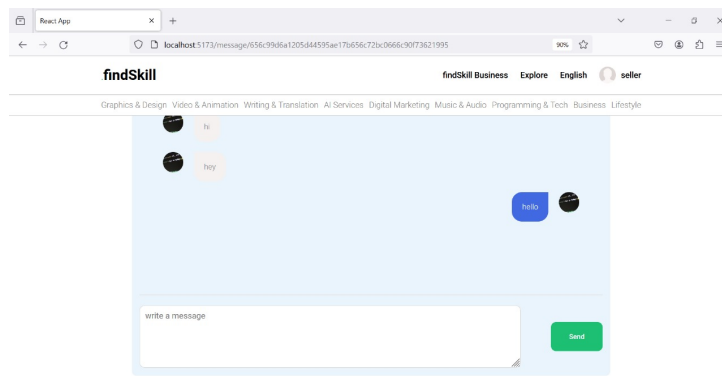


Figure 5.12: The chat page

CHAPTER 6

FUTURE SCOPE

The **FindSkill** project has several potential future scopes for further improvement and expansion:

1. Global Expansion: Expand the platform to cater to an international audience by integrating multi-language support, localized content, and consider regional payment gateways to enhance accessibility.

2. Skill Assessment and Certification: Provide a more detailed skill assessment and certification mechanism for freelancers by introducing skill tests, certification courses, and badges to showcase verified skills and expertise.

3. AI-driven Matching Algorithm: Enhance the project matching process using artificial intelligence by implementing machine learning algorithms to analyze user behavior, preferences, and project requirements for better matchmaking.

4. Smart Contracts for Transactions: Improve the transparency and security of transactions by exploring blockchain technology and smart contracts to automate and secure the financial aspects of project transactions.

5. Enhanced Collaboration Tools: Facilitate more seamless collaboration between freelancers and businesses by Integrate additional collaboration tools such as virtual whiteboards, project management tools, and real-time collaborative document editing.

6. Advanced Analytics Dashboard: Provide businesses and freelancers with in-depth analytics.

7. Community and Forum Features: Foster a sense of community among freelancers by creating discussion forums, interest groups, or community-driven features to facilitate networking and knowledge-sharing.

8. Voice and Video Integration: Enhance communication between freelancers and businesses by integrating voice and video calling features directly into the platform for real-time communication.

These future scopes represent potential directions for expanding and improving the FindSkill freelancer platform, providing more value to users and adapting to the evolving landscape of the freelancing industry. Prioritization should be based on user feedback, market trends, and technological advancements.

CHAPTER 7

CONCLUSION

In conclusion, the development and realization of FindSkill mark a transformative journey in the realm of freelancing platforms. This project, intricately designed using the MERN stack (MongoDB, Express.js, React, Node.js), is driven by a fundamental objective: to empower individuals to showcase their skills while enabling businesses to seamlessly connect with a diverse pool of talent. The user-friendly interface, underpinned by transparent communication channels, aspires to create an environment conducive to collaboration, innovation, and professional growth.

From a technical standpoint, the project's foundation in the MERN stack underscores its commitment to scalability, flexibility, and efficiency. Real-time communication adds a layer of dynamism to the platform, offering features like live chat and project updates. The frontend, constructed with React and styled using SCSS, places a premium on a responsive and intuitive design. React components contribute to modularity and reusability, fostering a consistent and seamless user experience. The backend, orchestrated by Node.js and Express.js, integrates seamlessly with MongoDB for data storage, with Mongoose serving as the intermediary for efficient database interactions. Security measures, including user authentication, data validation, and sanitization, underscore the project's commitment to safeguarding user data.

Looking forward, FindSkill holds promising avenues for future development. Concepts such as AI-driven matching algorithms, smart contracts for transactions, and integration with emerging technologies like AR/VR present exciting possibilities for innovation. The platform's scalability positions it favorably for global expansion and the incorporation of diverse payment options, further enriching its utility and reach.

Beyond its technical achievements, FindSkill aims to make a meaningful impact on the landscape of freelancing. By transcending traditional employment boundaries, it contributes to the evolving gig economy, providing a platform that fosters flexibility and dynamism in work arrangements. In essence, FindSkill is not just a technological achievement; it stands as a testament to the potential for positive change in the digital age. As the platform continues to evolve, it is poised to play a pivotal role in shaping the future of collaborative success in the freelancing landscape.

APPENDIX A

CODE ATTACHMENTS

A.1 App.jsx File

```
1 import './app.scss';
2 import { createBrowserRouter, Outlet, RouterProvider } from 'react-router-dom';
3 import React from 'react';
4 import Navbar from './components/navbar/Navbar';
5 import Footer from './components/footer/Footer';
6 import Home from './pages/home/Home';
7 import Gigs from './pages/gigs/Gigs';
8 import Gig from './pages/gig/Gig';
9 import Login from './pages/login/Login';
10 import Register from './pages/register/Register';
11 import Add from './pages/add/Add';
12 import Orders from './pages/orders/Orders';
13 import Messages from './pages/messages/Messages';
14 import Message from './pages/message/Message';
15 import MyGigs from './pages/myGigs/MyGigs';
16 import {
17   QueryClient,
18   QueryClientProvider,
19 } from '@tanstack/react-query';
20 import Pay from './pages/pay/Pay';
21 import Success from './pages/success/Success';
22 function App() {
23   const queryClient = new QueryClient();
24
25   const Layout = () => {
26     return (
27       <div className="app">
28         <QueryClientProvider client={queryClient}>
29           <Navbar />
30           <Outlet />
31           <Footer />
32         </QueryClientProvider>
33       </div>
34     );
35   };
36
37   const router = createBrowserRouter([
38     {
39       path: "/",
40       element: <Layout />,
41       children: [
42         {
43           path: "/",
44           element: <Home />,

```

```

45     },
46     {
47         path: "/gigs",
48         element: <Gigs />,
49     },
50     {
51         path: "/myGigs",
52         element: <MyGigs />,
53     },
54     {
55         path: "/orders",
56         element: <Orders />,
57     },
58     {
59         path: "/messages",
60         element: <Messages />,
61     },
62     {
63         path: "/message/:id",
64         element: <Message />,
65     },
66     {
67         path: "/add",
68         element: <Add />,
69     },
70     {
71         path: "/gig/:id",
72         element: <Gig />,
73     },
74     {
75         path: "/register",
76         element: <Register />,
77     },
78     {
79         path: "/login",
80         element: <Login />,
81     },
82     {
83         path: "/pay/:id",
84         element: <Pay />,
85     },
86     {
87         path: "/success",
88         element: <Success />,
89     },
90 ],
91 },
92 ]);
93
94 return <RouterProvider router={router} />;
95 }
96
97 export default App;

```

A.2 index.js File

```
1 // const express =require("express");
2 import express from "express";
3 import mongoose from "mongoose";
4 import dotenv from 'dotenv';
5 import userRoute from "../routes/user.route.js";
6 import gigRoute from "../routes/gig.route.js";
7 import orderRoute from "../routes/order.route.js";
8 import conversationRoute from "../routes/conversation.route.js";
9 import messageRoute from "../routes/message.route.js";
10 import reviewRoute from "../routes/review.route.js";
11 import authRoute from "../routes/auth.route.js";
12 import cookieParser from "cookie-parser";
13 import cors from "cors";
14
15 const app=express();
16 dotenv.config();
17 mongoose.set('strictQuery',true)
18 try {
19     await mongoose.connect(process.env.MONGO);
20 } catch (error) {
21     console.log(error);
22 };
23 app.use(cors({origin:"http://localhost:5173",credentials:true}))
24 app.use(express.json());
25 app.use(cookieParser());
26 app.use("/api/auth", authRoute);
27 app.use("/api/users", userRoute);
28 app.use("/api/gigs", gigRoute);
29 app.use("/api/orders", orderRoute);
30 app.use("/api/conversations", conversationRoute);
31 app.use("/api/messages", messageRoute);
32 app.use("/api/reviews", reviewRoute);
33 app.use((err,req,res,next)=>{
34     const errorStatus=err.status ||500
35     const errorMessage=err.message || "wrong"
36     return res.status(errorStatus).send(errorMessage);
37
38 });
39
40 app.listen(8800,()=>{
41     // connect();
42     console.log("running")
43 })
```

A.3 auth.controller File

```
1 import User from "../models/user.model.js";
2 import createError from "../utils/createError.js";
3 import bcrypt from "bcrypt";
4 import jwt from "jsonwebtoken";
5
```

```

6 export const register = async (req, res, next) => {
7   try {
8     const hash = bcrypt.hashSync(req.body.password, 5);
9     const newUser = new User({
10       ...req.body,
11       password: hash,
12     });
13
14     await newUser.save();
15     res.status(201).send("User■has■been■created.");
16   } catch (err) {
17     next(err);
18   }
19 };
20 export const login = async (req, res, next) => {
21   try {
22     // console.log("in■login■api")
23     const user = await User.findOne({ username: req.body.username });
24
25     if (!user) return next(createError(404, "User■not■found!"));
26
27     const isCorrect = bcrypt.compareSync(req.body.password, user.password);
28     if (!isCorrect)
29       return next(createError(400, "Wrong■password■or■username!"));
30
31     const token = jwt.sign(
32       {
33         id: user._id,
34         isSeller: user.isSeller,
35       },
36       process.env.JWT_KEY
37     );
38
39     const { password, ...info } = user._doc;
40     res
41       .cookie("accessToken", token, {
42         httpOnly: true,
43       })
44       .status(200)
45       .send(info);
46   } catch (err) {
47     next(err);
48   }
49 };
50
51 export const logout = async (req, res) => {
52   res
53     .clearCookie("accessToken", {
54       sameSite: "none",
55       secure: true,
56     })
57     .status(200)
58     .send("User■has■been■logged■out.");
59 };

```

A.4 user.model File

```
1 import mongoose from "mongoose";
2 const { Schema } = mongoose;
3
4 const userSchema = new Schema({
5   username: {
6     type: String,
7     required: true,
8     unique: true,
9   },
10  email: {
11    type: String,
12    required: true,
13    unique: true,
14  },
15  password: {
16    type: String,
17    required: true,
18  },
19  img: {
20    type: String,
21    required: false,
22  },
23  country: {
24    type: String,
25    required: true,
26  },
27  phone: {
28    type: String,
29    required: false,
30  },
31  desc: {
32    type: String,
33    required: false,
34  },
35  isSeller: {
36    type: Boolean,
37    default: false
38  },
39 }, {
40   timestamps: true
41 });
42
43 export default mongoose.model("User", userSchema)
```

A.5 auth.route File

```
1 import express from 'express';
2 import { register, login, logout } from '../controllers/auth.controller.js';
3 const router = express.Router();
4 router.post("/register", register);
5 router.post("/login", login);
```

```

6 router.post("/logout",logout)
7 export default router;

```

A.6 login.jsx File

```

1 import React, { useState } from "react";
2 import "../Login.scss";
3 import newRequest from "../../utils/newRequest";
4 import { useNavigate } from "react-router-dom";
5 import axios from "axios";
6
7 function Login() {
8   const [username, setUsername] = useState("");
9   const [password, setPassword] = useState("");
10  const [error, setError] = useState(null);
11
12  const navigate = useNavigate();
13
14  const handleSubmit = async (e) => {
15    e.preventDefault();
16    try {
17      // console.log("Button")
18      const res=await newRequest.post("/auth/login", { username, password
19        });
20      localStorage.setItem("currentUser", JSON.stringify(res.data));
21      navigate("/")
22    } catch (err) {
23      setError(err.response.data);
24    }
25  };
26
27  return (
28    <div className="login">
29      <form onSubmit={handleSubmit}>
30        <h1>Sign in</h1>
31        <label htmlFor="">Username</label>
32        <input
33          name="username"
34          type="text"
35          placeholder="johndoe"
36          onChange={(e) => setUsername(e.target.value)}
37        />
38        <label htmlFor="">Password</label>
39        <input
40          name="password"
41          type="password"
42          onChange={(e) => setPassword(e.target.value)}
43        />
44        <button type="submit">Login</button>
45        {error && error}
46      </form>
47    </div>);}
48 export default Login;

```

CHAPTER 8

BIBLIOGRAPHY

MongoDB Documentation: [Click here to access](#)

Express.js Documentation: [Click here to access](#)

React Documentation: [Click here to access](#)

Node.js Documentation: [Click here to access](#)

Vite Documentation: [Click here to access](#)

Understanding the Basics of RESTful API Design: [Click here to access](#)

React Router Documentation [Click here to access](#)

8.0.1 Project Link of FindSkill:

[Go to Github](#)